



Rubric Design 101

How to Design Rubrics 101

Written by one of our talented Super Reviewers, this is a beginner-friendly guide to rubrics for this project! We encourage all new writers to read this section.

Before we dive into how to design rubrics for Agentic Code QA, let us step back to basics. Please note that **the first three sections apply to any AI training dataset** and not this project alone. It's a universal guide.

What are Rubrics

Rubrics are **evaluation parameters**. If you have ever done an assignment that has multiple right answers, **think of Rubrics as the evaluation parameters for the best possible right answer (our golden answer)**. That said, the other answers are not wrong, they are just not the best. **Agentic Code QA is looking for rubrics for the best possible answer.**

How are Rubrics Used

When a model is evaluated for correctness and other parameters, the model is expected to pass the rubric evaluation with a score of >85% (the threshold for “too easy”). If the model fails to do so, then it requires further training. In the AI world, this is called Reinforcement Learning. **You are teaching the model by rewarding it for correctness and accuracy and penalizing it for wrongdoing.** Just like how we teach children.

How to Write Rubrics

You have to first **stop thinking like a developer** and **start thinking like a parent/teacher**. Imagine you are coaching a child to provide your Golden Answer and have to evaluate them. What would you do? Ask them questions, give assessments to test their ability to recollect the answers, etc. Rubrics do the same thing. If something is marked `critical` in a rubric and the model does not do that, your model is not performing well. As discussed in the [section on how to make a task Hard](#), your answer has to be the best and most efficient answer without following any shortcuts. That makes rubrics vital to evaluate how good your answer is and helps in rewarding a model that produces it. AI training companies then use this to retrain their models by updating their weights accordingly.

Rubrics in Agentic Code QA

Rubrics are divided into 3 tags and 3 weights. We shall go through each one of them one by one and understand how and when to use them. By the end, I will teach you how to frame the rubrics in your mind while thinking about the golden answer.

Tags

1. **Reasoning:** Every question needs thinking. **You cannot answer a question without thinking.** Even if someone asks you 1×1 and you answer 1, your brain has thought about it, although for a split second. This thinking is what goes into Reasoning. All the thinking parts your golden answer is going to have, like thinking about using a piece of code, identifying something in the code, etc., are all part of Reasoning rubrics. All the statements that lead to the final answer are also Reasoning. **You are giving the model a step-by-step thinking methodology to arrive at your golden answer.** You are telling the model how to get there properly.
2. **Completeness:** Suppose your Golden Answer said something that completes the answer, say it has found **something that is important for understanding but it was discovered as part of the thinking process, that is completeness.** You can always use this to differentiate between Reasoning and Completeness. If it is important for thinking, it is tagged Reasoning. If it was discovered *as part of the thinking but needed*, it is tagged Completeness.
3. **Style:** This is the easiest of all. Note that Style *cannot be critical*. That is because you cannot penalize a model for not following one style. If I answer “ 1×1 is 1” or “1 is the answer for 1×1 ”, both are correct. This is style. **The presentation of the answer, how the parts and parcels of the code are used, how code is displayed, all these come under style.**

Weight

1. **Critical:** This is a **must-have** in the model's answer. **The model fails terribly if it does not generate this.** The model must be retrained if this is not answered. For example, imagine teaching a child about how humans breathe and they forget to mention or think about the nose. Would you accept that? Would you retrain the child by telling them that they also have to think about the nose?
2. **Bonus:** This is good to have. **The model does not fail, but it is good to have.** For example, the child says that we breathe through our mouth when the nose is blocked. This is a good thing to have in the answer. The answer is **not wrong without it, but is better with it.**

3. **Penalty:** This should *not* be in the model's answer. **You are giving negative criteria so the model stops doing this.** For example, if the child says that we can breathe through our ears, this is absolutely wrong and not possible, but perhaps the child is very persistent. You correct them, but first you penalize them for saying this *because you know it is incorrect* and would be a bad way for the child to continue thinking.

With this in mind, first try to get to the Critical rubric criteria required in the answer. Forget about bonus or penalty for a minute and just focus on Critical. **Note down all the rubric criteria that are required for the model to succeed while writing the golden answer itself.** If you use an LLM, then you would be disqualified. **Do not use an LLM to write your rubrics.**

If you write two lines in the golden answer that can form one rubric criterion, write that in parallel. You also evaluate your answer at the same time.

After all your Critical rubric criteria are covered, then check your Golden Answer. Did you mention something that is a Bonus? Include that. Then and only then, go ahead and write the Style criteria. See how your answer was framed, what was shown, etc. That should give you some basics.

Then go to the Penalties. Check rules for penalties below, but think of it like: *the model must not do this*. You frame these as if the model did this bad thing that it was not supposed to do. To write this, **imagine you are an evil developer who is asking a model to solve a prompt. Yup, evil. Apply your evil skills.**

Your rubric tells how to answer your prompt. If your prompt is already good enough and your answer has extra points/does not follow any shortcuts, you should have the right number of rubric criteria, no matter what.

Rules to Write Rubrics

I won't go into the basics here (basic rubric writing is covered elsewhere), and will focus on Hard tasks only.

1. Your rubric must award at least one critical point for an alternative answer that is correct. **The model thought a different way than you? No problem. That is not wrong.** At least give a mark for its efforts.

2. Rubric criteria must not overlap. **One rubric criterion must cover one part of the Golden Answer only.** That being said, you can combine a part or two of the golden answer into one rubric criterion, but you must follow Point 5 below, so be careful on what you are combining.
3. **Rubric criteria must be specific, but not overly specific.** Do not mention in detail the exact file, line, name, etc. Mentioning a function name in so-and-so file is OK, but going in depth is not acceptable. Why not? *You would force the model to answer your way.* This is not correct, because it closes off the ability to answer (or think) differently than you.
4. Rubrics **must not be vague or use vague/non-quantified terms. No words like “correctly,” “incorrectly,” “easy,” “clear,” etc.** Every rubric criterion is, by definition, a test of correctness. Avoid any subjective word like “good” or “clear” because it has no universal definition.
5. **Rubrics must be atomic.** One rubric criterion must cover one and only one part of the golden answer. Anything more will get rejected. There are exceptions, but those will be discussed on a case by case basis. Try to avoid “and” and “or” in your rubrics as much as you can. (The exceptions are usually when two things are inextricably linked, you cannot *ever* have one without the other, so it is OK to test them together in one rubric criterion. This is rare, but it does happen. A good case for using “or” is when there is a clearly scoped set of right answers, and *any one of them is OK*. In that case you should enumerate all of them and write something like: the model must do a, b, or c.)
6. **Penalties cannot be direct reversals of critical and bonus rubrics.** That would not make sense because you have already awarded points for doing the positive behavior. If you also took away points for *not doing the positive behavior*, you would be “double counting,” meaning penalizing the same behavior twice.

So the main points to remember are: **think like you are teaching a child how to answer a question** and **rubric criteria are used to evaluate the answer**. Get into the boots of a parent or teacher and writing rubrics will get very easy.